Extra study and consolidation in Algebra

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1 What is this?

Due to my being very interested in algebra, and interested in becoming very good at it, I have these extra notes for me to work through a lot of problems. Basically, there will probably not be any pedagogical value here.

2 Repeating some stuff about ideals, exact sequences and so forth

Problem 2.1. Let

$$R = \frac{k[x,y]}{(xy)}$$

and write $\overline{x}, \overline{y}$ for the residue classes.
What is $(\overline{x})$ as an ideal of R?
Compute

$$R/(\bar{x}), \qquad R/(\bar{y}), \qquad R/(\bar{x},\bar{y}).$$

Then classify them as prime/maximal/neither.

Sol. We note that the ideal generated by $\overline{x} = x + (xy)$ is

$$(\overline{x}) = \{\overline{xf} \mid f \in k[x,y]\}.$$

We have $(xy) \subseteq (x)$ and $(xy) \subseteq (y)$ as ideals of $k[x,y]$.

Consider the projection $\pi : k[x,y] \to R$ by

$$\pi(a) = \overline{a} = a + (xy).$$

Since π is surjective and $\{x\} \subseteq k[x,y]$ we have

$$\pi[(x)] = (\pi x) = (\overline{x}),$$

(note: surjectivity guarantees that scalar multiplication in R can be pulled back to $k[x,y]$, ensuring $\pi[(x)] \supseteq (\pi(x))$) but also

$$\pi[(x)] = \frac{(x)}{(xy)}$$

since $(xy) \subseteq (x)$.

Similarly,

$$\frac{(y)}{(xy)} = \pi[(y)] = (\overline{y}),$$

and

$$\frac{(x,y)}{(xy)} = \pi[(x,y)] = (\overline{x},\overline{y}).$$

Thus we can apply the third isomorphism theorem to get

$$\frac{R}{(\overline{x})} = \frac{k[x,y]/(xy)}{(x)/(xy)} \cong \frac{k[x,y]}{(x)} \cong k[y],$$

and

$$\frac{R}{(\overline{y})} = \frac{k[x,y]/(xy)}{(y)/(xy)} \cong \frac{k[x,y]}{(y)} \cong k[x],$$

and

$$\frac{R}{(\overline{x},\overline{y})} = \frac{k[x,y]/(xy)}{(x,y)/(xy)} \cong \frac{k[x,y]}{(x,y)} \cong k.$$

The first two are integral domains so $(\overline{x}), (\overline{y})$ are prime, though not maximal, while $(\overline{x},\overline{y})$ is maximal since $R/(\overline{x},\overline{y}) \cong k$ is a field. ■

Problem 2.2. Let

$$R = \frac{k[x,y]}{(xy)},$$

and let

$$\pi : k[x,y] \to R$$

be the quotient map.
Prove that every prime ideal $P \subseteq R$ contains $\overline{x}$ or $\overline{y}$.
Then, classify every prime ideal of R.
You may use without proof:

$$P \subseteq R \text{ prime} \Leftrightarrow \pi^{-1}(P) \subseteq k[x,y] \text{ is prime containing } (xy).$$

Proof. We have $\overline{xy} = 0$, and since P is an ideal, and thus a group, we have $0 \in P$, and since P is prime we must then have

$$\overline{x} \in P \text{ or } \overline{y} \in P.$$

Suppose P contains $\overline{x}$. Then, $(\overline{x}) \subseteq P$.

Then, by the third isomorphism theorem,

$$\frac{R/(\overline{x})}{P/(\overline{x})} \cong R/P,$$

but

$$\frac{R}{(\overline{x})} \cong k[y]$$

so

$$R/P \cong \frac{k[y]}{P/(\overline{x})}.$$

Since P is a prime ideal of R, the quotient R/P is an integral domain. Therefore, $P/(\overline{x})$ must be a prime ideal of $k[y]$.

Since $k[y]$ is a Principal Ideal Domain (PID), its prime ideals are:

1. The zero ideal (0), or
2. Irreducible polynomial ideals $(f(y))$, where $f(y) \in k[y]$ is an irreducible polynomial.

Under the canonical quotient map $R \to R/(\overline{x}) \cong k[y]$, these correspond precisely to the prime ideals of R containing $\overline{x}$:

1. $P = (\overline{x})$, corresponding to $(0) \subset k[y]$.
2. $P = (\overline{x}, f(\overline{y}))$, where $f(y) \in k[y]$ is irreducible.

By symmetric reasoning, if $\overline{y} \in P$, then $P/(\overline{y})$ is a prime ideal of $k[x]$, which yields:

1. $P = (\overline{y})$, corresponding to $(0) \subset k[x]$.
2. $P = (\overline{y}, g(\overline{x}))$, where $g(x) \in k[x]$ is irreducible.

Combining both cases, the prime ideals of R consist of:

- Minimal prime ideals: $(\overline{x})$ and $(\overline{y})$.
- Maximal ideals: $(\overline{x}, f(\overline{y}))$ for irreducible $f(y) \in k[y]$, and $(\overline{y}, g(\overline{x}))$ for irreducible $g(x) \in k[x]$.

■

Problem 2.3. Let R be a commutative ring, I an ideal of R, and M an R-module. You have three objects

$$IM,\ M,\ M/IM.$$

Part A.
Identify the two most natural R-linear maps between these objects and use them to construct a short exact sequence.
Give the maps explicitly and check exactness at every nonzero term.
Part B.
Now consider R/I as an R-module.
There is a canonical isomorphism

$$M \otimes_R R/I \cong M/IM.$$

Answer:

- What should the natural map

$$M \times R/I \to M/IM$$

  probably be?
- Why is there an immediate well-definedness issue?
- What property would allow that bilinear map to induce a map from

$$M \otimes_R R/I?$$

Part C.
Suppose $N \subseteq M$. There is an obvious short exact sequence

$$0 \to N \to M \to M/N \to 0.$$

Tensor all with R/I.
You might hope to get

$$0 \to N \otimes_R R/I \to M \otimes_R R/I \to (M/N) \otimes_R R/I \to 0.$$

Is that always exact?

Proof of A. IM is the collection of finite sums of $i_j m_j$ where $i_j \in I$ and $m_j \in M$.

M/IM is then the collection of

$$\overline{x} = x + IM,\ x \in M,$$

meaning the residue classes of elements in M modulo IM.

The most natural map from M to M/IM is the projection $\pi : M \to M/IM$,

$$\pi(x) = \overline{x}$$

sending elements x into their residue classes.

The natural map from IM to M is the inclusion $\iota : IM \to M$ simply sending

$$\iota(x) = x$$

since $IM \subseteq M$.

R-linearity of ι is essentially trivial since

$$\iota(rx + y) = rx + y = r\iota(x) + \iota(y).$$

For π observe that

$$\begin{aligned}\pi(rx+y) &= \overline{rx+y} \\ &= (rx+y)+IM \\ &= (rx+IM)+y+IM \\ &= r(x+IM)+(y+IM) \\ &= r\overline{x}+\overline{y} \\ &= r\pi(x)+\pi(y).\end{aligned}$$

This gives a natural sequence

$$0 \to IM \xrightarrow{\iota} M \xrightarrow{\pi} M/IM \to 0.$$

Then

$$\ker \pi = IM$$

since elements in IM are 0 modulo IM when you take the quotient, and $IM \subseteq M$.

Then

$$\operatorname{im} \iota = IM,$$

since ι maps x to x with $IM \subseteq M$.

We have $\operatorname{im} \pi = M/IM$ since a residue class $\overline{x} \in M/IM$ has at least one representative $x \in M$.

The only map from $M/IM \to 0$ is the trivial map, so

$$\ker(M/IM \to 0) = M/IM.$$

For the other side of the sequence there is once again only the trivial map from $0 \to IM$ with $\operatorname{im}(0 \to IM) = 0$, and

$$\ker \iota = 0,$$

since $\iota(x) = 0_M$ implies $x = 0_M$, but as $IM \subseteq M$ we have that $0_{IM} = 0_M$.

Thus we have shown exactness at each point in the sequence, so this short sequence is exact. ■

Proof of B. A natural map $\beta : M \times R/I \to M/IM$ might be

$$(m, \overline{x}) \xmapsto{\beta} \overline{mx},$$

meaning

$$(m, r+I) \mapsto (mr)+IM$$

Notice that $(m, \overline{x}) \mapsto \overline{m+x}$ would very obviously create issues if we try to apply it to the tensor product since we require that $m \otimes 0$ is 0. Furthermore, $m+r$ is generally not an

element of M for arbitrary $r \in R$.

For well-definedness suppose $r + I = s + I$. Then $r - s \in I$.

Then $(r - s)m \in IM$ since $r - s \in I$ and $m \in M$.

Thus we have that

$$\begin{aligned}\beta(m, r) - \beta(m, s) &= (mr + IM) - (ms + IM) \\ &= (m(r - s)) + IM,\end{aligned}$$

bit $m(r - s) \in IM$ so this is 0_{IM}.

Lastly

$$\begin{aligned}\beta(m_1 + m_2, \overline{r}) &= ((m_1 + m_2)r) + IM \\ &= (m_1 r + m_2 r) + IM \\ &= (m_1 r + IM) + (m_2 r + IM) \\ &= \beta(m_1, \overline{r}) + \beta(m_2, \overline{r}),\end{aligned}$$

and

$$\begin{aligned}\beta(m, \overline{r} + \overline{s}) &= (m(r + s)) + IM \\ &= (mr + ms) + IM \\ &= (mr + IM) + (ms + IM) \\ &= \beta(m, \overline{r}) + \beta(m, \overline{s}),\end{aligned}$$

and

$$\begin{aligned}\beta(am, \overline{r}) &= ((am)r) + IM \\ &= (m(ar)) + IM \\ &= \beta(m, a\overline{r}).\end{aligned}$$

Thus, by the universal property of the tensor product, we can factor through a unique map so that

$$\begin{array}{ccc} M \times R/I & \xrightarrow{\beta'} & M/IM \\ & \searrow & \uparrow{\scriptstyle \exists!} \\ & & M \otimes_R R/I \end{array}$$

commutes. ■

Proof of C. We begin with the short exact sequence

$$0 \to N \xrightarrow{\iota} M \xrightarrow{\pi} M/N \to 0.$$

Tensoring with R/I induces maps $\iota_R : N \otimes_R R/I \to M \otimes_R R/I$ sending

$$n \otimes \overline{r} \overset{\iota_R}{\mapsto} n \otimes \overline{r}$$

by

$$\iota_R(n \otimes \overline{r}) = \iota(n) \otimes r,$$

and $\pi_R : M \otimes_R R/I \to (M/N) \otimes_R R/I$ sending

$$m \otimes \overline{r} \overset{\pi_R}{\mapsto} \overline{m} \otimes \overline{r}$$

by

$$\pi_R(m \otimes \overline{r}) = \pi(m) \otimes \overline{r} = (m + N) \otimes \overline{r}.$$

In general tensoring is right exact, meaning

$$M' \otimes L \to M'' \otimes L \to M''' \otimes L \to 0$$

would remaing exact if

$$M' \to M'' \to M''' \to 0$$

was.

Take for instance

$$0 \to 2\mathbb{Z} \to \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to 0$$

and tensor with $\mathbb{Z}/2\mathbb{Z}$.

Then we imagine we get an exact sequence

$$0 \to 2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \to 0.$$

However, there is no such exact sequence, with the

$$0 \to 2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z}$$

term being the problem.

Consider $2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2 \to \mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2$. Take $2 \otimes \overline{1}$. Then the image is $2 \otimes \overline{1}$, but in $\mathbb{Z} \otimes \mathbb{Z}/2$ we have

$$(2 \cdot 1) \otimes \overline{1} = 1 \otimes (2\overline{1}) = 0.$$

In $2\mathbb{Z} \otimes \mathbb{Z}/2$, however, $2 \otimes \overline{1}$ is nonzero. Take the $\mathbb{Z}$-bilinear map $B : 2\mathbb{Z} \times \mathbb{Z}/2 \to \mathbb{Z}_2$ given by $B(2m, \overline{n}) = mn \pmod 2$ and take the unique map factoring through the tensor product $\beta : 2\mathbb{Z} \otimes \mathbb{Z}/2 \to \mathbb{Z}_2$.

As an asside, the map B is well defined since taking the product of two integers modulo 2 gives an element in $\mathbb{Z}/2\mathbb{Z} \cong \mathbb{Z}_2$.

It is linear since

$$\begin{aligned} B(2m+2m',n) &= (2m+2m')n \mod 2 \\ &= 2mn+2m'n \mod 2 \\ &= B(2m,n)+B(2m'n) \end{aligned}$$

in $\mathbb{Z}/2\mathbb{Z}$. The argument for linearity in the second coordinate and for scalar multipliaction from $\mathbb{Z}$ on the coordinates is analogous.

Then

$$\beta(2\otimes 1) = 1 \neq 0.$$

Thus there is a nonzero element sent to 0, so

$$\ker(\iota_R) \neq 0.$$

■

Problem 2.4. Let M be an R-module and let $A, B \leq M$ be submodules. You are given

$$A\cap B,\ A,\ B,\ A+B.$$

Construct a natural short exact sequence involving all.

Sol. We note that

$$A+B \supseteq A \supseteq A\cap B$$

and similarly,

$$A+B \supseteq B \supseteq A\cap B.$$

Thus there are very natural inclusion maps

$$A\cap B \to A,\ A\cap B\to B,\ A\to A+B,\ B\to A+B.$$

However, these are not what we are looking for exactly.

The observation that

$$\frac{A+B}{B} \cong \frac{A}{A\cap B}$$

gives us what we need.

Let $\iota : A\cap B \to A$ be the inclusion, which is well-defined since $A\cap B \subseteq A$.

Then define $\pi : A \to (A+B)/B$ to be the map

$$\pi(a) = a+B.$$

This feels like the projection map since for $a+b$ in $A+B$ we have $(a+b)+B = a+B$ since $b\in B$.

It is not terribly hard to see that these are homomorphisms.

Then

$$\operatorname{im}(0 \to A \cap B) = 0 = \ker \iota,$$

and

$$\operatorname{im} \iota = A \cap B = \ker \pi$$

since the $A \cap B \subseteq A$ go to 0 modulo B, and

$$\operatorname{im} \pi = \frac{A+B}{B} = \ker(A + B/B \to 0),$$

since π is surjective: take $(a+b)+B \in (A+B)/B$, then

$$(a+b)+B = a+B$$

since $b \in B$, which is mapped to by $a \in A$.

Thus

$$0 \to A \cap B \xrightarrow{\iota} A \xrightarrow{\pi} \frac{A+B}{B} \to 0$$

is exact, reaffirming the fact that, for exact sequence

$$0 \to A \to B \to C \to 0$$

we have

$$C \cong \frac{B}{\operatorname{im}(A \to B)}.$$

Another natural route is to look at $\psi : A \oplus B \to A + B$ by

$$\psi(a,b) = a + b.$$

Then the kernel forces

$$\begin{aligned} a + b &= 0 \\ a &= -b \end{aligned}$$

so $a, b \in A \cap B$. This points to the natural map $\varphi : A \cap B \to A \oplus B$ by

$$\varphi(x) = (x, -x).$$

Then the sequence

$$0 \to A \cap B \xrightarrow{\varphi} A \oplus B \xrightarrow{\psi} A + B \to 0$$

is exact.

This is not hard to prove as

$$\operatorname{im}(0 \to A \cap B) = 0 = \ker \varphi$$

since $(x, -x) = 0$ iff $x = 0$. Then

$$\operatorname{im}\varphi = \{(x, -x) \mid x \in A \cap B\} = \ker\psi$$

by the earlier argument. Lastly

$$\operatorname{im}\psi = A + B = \ker(A + B \to 0)$$

since ψ is surjective (for $a + b \in A + B$ there is $(a, b) \in A \oplus B$). ■

Problem 2.5. Let M be an R-module and let $r \in R$. Consider only

$$M,\ \operatorname{Ann}_M(r) = \{m \in M \mid rm = 0\},\ M/rM.$$

There is a very natural endomorphism

$$M \to M$$

associated with r.
Discover an exact sequence involving

$$\operatorname{Ann}_M(r),\ M,\ M,\ M/rM.$$

Solution and proof. A natural endomorphism on M associated with r could be $\rho : M \to M$ by

$$\rho(m) = rm.$$

Since modules are closed under multiplication from the ring, this is well defined. It is not terribly hard to see that it is indeed a homomorphism since

$$\begin{aligned}\rho(am_1 + m_2) &= r(am_1 + m_2)\\ &= ram_1 + rm_2\\ &= a(rm_1) + rm_2\\ &= a\rho(m_1) + \rho(m_2).\end{aligned}$$

This map need not be injective, and it need not be surjective either. Take for instance R to be $\mathbb{Z}$ and M to be $\mathbb{Z}/2\mathbb{Z}$ with $r = 2$. Then the image does not include $\overline{1}$.

The first useful fact is that for $m \in \operatorname{Ann}_M(r) \subseteq M$ we have $m \overset{\rho}{\mapsto} 0$.

Which gives a candidate:

$$0 \to \operatorname{Ann}_M(r) \overset{\iota}{\to} M \overset{\rho}{\to} M \overset{\pi}{\to} M/rM \to 0.$$

Let $\iota : \operatorname{Ann}_M(r) \to M$ be the inlcusion map, which is well-defined since $\operatorname{Ann}_M(r) \subseteq M$.

Then let $\pi : M \to M/rM$ be the projection map

$$m \overset{\pi}{\mapsto} \overline{m}.$$

Then

$$\text{im}(0 \to \text{Ann}_M(r)) = 0_{\text{Ann}_M(r)},$$

but $\text{Ann}_M(r) \subseteq M$ so

$$0_{\text{Ann}_M(r)} = 0_M,$$

hence

$$\text{im}(0 \to \text{Ann}_M(r)) = \ker \iota.$$

Next, we have that

$$\text{im}\, \iota = \text{Ann}_M(r).$$

By definition, for $m \in M$,

$$mr = \rho(m) = 0$$

if and only if

$$m \in \text{Ann}_M(r),$$

so

$$\ker \rho = \text{Ann}_M(r) = \text{im}\, \iota.$$

Next

$$\text{im}\, \rho$$

is $rM = \{rm \mid m \in M\}$, but this is just kernel $\ker \pi$ since $x = 0$ in M/rM if and only if $x \in rM$ if and only if

$$x = rm,\ m \in M.$$

Thus

$$\text{im}\, \rho = \ker \pi.$$

Lastly,

$$\text{im}\, \pi = M/rM$$

as the projection is always surjective sicne there is at least one representative $m \in M$ for each residue class modulo rM.

Now we take a look at a concrete example with $R = \mathbb{Z}$, $M = \mathbb{Z}/12\mathbb{Z}$ and $r = 8$.

Then

$$\text{Ann}_M(8) = \{\overline{m} \in \mathbb{Z}/12\mathbb{Z} \mid m \cdot 8 = n \cdot 12,\ n \in \mathbb{Z}\}.$$

. The annihilators (worked out concretely further down) are then

$$\overline{0},\ \overline{3},\ \overline{6},\ \overline{9},$$

meaning

$$\text{Ann}_M(8) = (3).$$

Then

$$8M$$

is the residue classes modulo 12 after being multiplied by $8 = 2 \cdot 2 \cdot 2$. Concretely,

$$\begin{aligned}
8 \cdot \overline{0} &= \overline{0} \\
8 \cdot \overline{1} &= \overline{8} \\
8 \cdot \overline{2} &= \overline{4} \\
8 \cdot \overline{3} &= \overline{0} \\
8 \cdot \overline{4} &= \overline{8} \\
8 \cdot \overline{5} &= \overline{4} \\
8 \cdot \overline{6} &= \overline{0} \\
8 \cdot \overline{7} &= \overline{8} \\
8 \cdot \overline{8} &= \overline{4} \\
8 \cdot \overline{9} &= \overline{0} \\
8 \cdot \overline{10} &= \overline{8} \\
8 \cdot \overline{11} &= \overline{4},
\end{aligned}$$

meaning

$$8M = (4)$$

in this scenario.

Note: We could also have worked out the annihilator by observing that

$$8m \cong 0(\mod 12)$$

if and only if

$$12 \mid 8m$$

if and only if

$$3 \mid 2m$$

if and only if (because 2 and 3 are coprime)

$$3 \mid m.$$

For the maps we have that the 0 map has image $\overline{0}$, with the kernel of the inclusion from the annihilator being $\overline{0}$ as well.

Furthermore the image of the inclusion, which I will here also denote ι is itself the annihilator

$$(3).$$

The kernel of the endomorphism sending $m \mapsto 8m$ is the elements for which $8m$ is 0,

namely

$$\overline{0},\ \overline{3},\ \overline{6},\ \overline{9}.$$

The image of the endomorphism is

$$8M = (4)$$

and the kernel of the projection π is $8M$. The image of the projection is the whole module $M/8M$ so the last term

$$\xrightarrow{\pi} M/8M \to 0$$

is exact.

We have that

$$\frac{\mathbb{Z}/12\mathbb{Z}}{(4)} \cong \mathbb{Z}/4\mathbb{Z}.$$

You can show this by taking the surjective homomorphism $\varphi : \mathbb{Z}/12\mathbb{Z} \to (4)$ by

$$\varphi(\overline{n}_{12}) = \overline{n}_4$$

and applying the first isomorphism theorem.

For well-definedness we observe that if $n_1 \equiv n_2$ modulo 12 then, since $4 \mid 12$, we have that $n_1 \equiv n_2$ modulo 4.

Is is a homomorphism since

$$\begin{aligned}\varphi(r\overline{n}_{12} + \overline{m}_{12}) &= r\overline{n}_4 + \overline{m}_4 \\ &= r\varphi(\overline{n}_{12} + \overline{m}_{12}).\end{aligned}$$

For surjectivity, observe that

$$\overline{0} \mapsto \overline{0},\ \overline{1} \mapsto \overline{1},\ \overline{2} \mapsto \overline{2},\ \ \overline{3} \mapsto \overline{3}.$$

Note: We surpress the earlier used subscripts denoting which modulo we're taking, but the point is it's obvious.

Then we get an exact sequence

$$0 \to (3) \xrightarrow{\iota} \mathbb{Z}/12\mathbb{Z} \xrightarrow{\times 8} \mathbb{Z}/12\mathbb{Z} \xrightarrow{\pi\circ\varphi} \mathbb{Z}/4\mathbb{Z} \to 0.$$

■

3 Switching over to more ideal-centric stuff and ties to algebraic geometry

Problem 3.1. Let

$$R = k[x, y, z],$$

and consider

$$I = (x^2y, x^2z^2),\ Q_1 = (x^2),\ Q_2 = (y, z^2).$$

(a) Compute

$$\sqrt{Q_1} \text{ and } \sqrt{Q_2}.$$

(b) Prove that both Q_1 and Q_2 are primary ideals. You may use the characterization that Q is primary if and only if every zero divisor of R/Q is Nilpotent.

(c) Prove that

$$I = Q_1 \cap Q_2.$$

You may **not** simply cite a general formula for intersections of monomial ideals. Use an algebraic argument.

(d) Deduce that

$$I = Q_1 \cap Q_2$$

is a primary decomposition. Determine whether it is irredundant and identify the corresponding prime ideals.

(e) Let

$$\mathfrak{p} = (x),\ \mathfrak{q} = (y, z).$$

Compute $I_{\mathfrak{p}}$ and $I_{\mathfrak{q}}$. Explain algebraically why one of the primary coponents dissappears under each localization.

(f) Finally, translate the decomposition into geometry by describing

$$V(I) \subseteq k^3.$$

Explain how the irreducible pieces you see are reflected in $\sqrt{Q_1}$ and $\sqrt{Q_2}$.

Solution to (a). We claim the radicals are

$$\sqrt{Q_1} = (x)$$

and

$$\sqrt{Q_2} = (y, z).$$

The backward inclusion $(x) \subseteq \sqrt{Q_1}$ is straightforward.

For $f \in (x)$ we have $f = ax$. Then $f^2 = a^2x^2 \in Q_1$ so $f \in \sqrt{Q_1}$.

Now suppose $f \in \sqrt{Q_1}$. For simplicity assume f is nonzero.

Then there exists $n \in \mathbb{N}$ such that

$$f^n = ax^2 \in Q_1 \subseteq (x),$$

Since (x) is prime we have that f must be in (x).

Thus $\sqrt{Q_1} = (x)$.

Similarly, we obviously have

$$(y, z) \subseteq \sqrt{Q_2}$$

since for $f = ay + bz \in (y, z)$ we can pick $n = 2$ and get

$$\begin{aligned} f^2 &= (ay + bz)(ay + bz) \\ &= a^2y^2 + 2ab(yz) + b^2z^2 \\ &= (a^2y + 2abz)y + (b^2)z^2 \in (y, z^2) = Q_2. \end{aligned}$$

Now suppose $f \in \sqrt{Q_2}$ so there exists $n \in \mathbb{N}$ such that

$$f^n = ay + bz^2 = ay + (bz)z \in (y, z).$$

■

Proof of (b). For

$$\frac{R}{Q_1} = \frac{k[x, y, z]}{(x^2)}$$

we have that any x term with degree ≥ 2 is 0.

Now suppose

$$\overline{f}\,\overline{g} = 0,$$

with both nonzero. Equivalently, suppose $f, g \in k[x, y, z]$ with $f, g \notin (x^2)$.

Then we have

$$f(x, y, z)g(x, y, z) = ax^2.$$

Both polynomials must be a multiple of x, i.e.

$$f(x, y, z) = x(q(x, y, z)).$$

Then

$$f^2 \in (ax^2).$$

Now for

$$\frac{R}{Q_2} = \frac{k[x, y, z]}{(y, z^2)} \cong \frac{k[x, z]}{(z^2)},$$

the argument here is entirely analogous.

Since R/Q_i for $i = 1, 2$ has that all zero divisors are nilpotent, we have that Q_1, Q_2 are primary. ■

Proof of (c) and (d). Suppose $f \in (x^2y, x^2z^2)$. Then

$$f = ax^2y + bx^2z^2.$$

Then

$$\begin{aligned} f &= ax^2y + bx^2z^2 \\ &= x^2(ay + (bx)z^2) \end{aligned}$$

which is in x^2 since it is divisible by x^2. Furthermore the term $ay + (bx)z^2$ is in (y, z^2).

Now suppose $f \in Q_1 \cap Q_2$. Then

$$f = ax^2$$

and

$$f = by + cz^2.$$

It is not hard to see that if

$$ax^2 = by + cz^2$$

then we require $a \in (y, z^2)$ and $b, c \in (x^2)$ so

$$a = a_1y + a_2z^2,$$

and

$$b = b_1x^2,\ c = c_1x^2.$$

Then it clearly follows that

$$\begin{aligned} f &= (a_1y + a_2z^2)x^2 + (b_1x^2)y + (c_1x^2)z^2 \\ &= (a_1b_1)x^2y + (a_2c_1)z^2x^2 \in (x^2y, x^2z^2). \end{aligned}$$

Thus

$$Q_1 \cap Q_2 \subseteq I$$

and

$$I = Q_1 \cap Q_2.$$

Thus, since Q_1, Q_2 are primary,

$$I = Q_1 \cap Q_2$$

is a primary decomposition, with $\sqrt{Q_1} = (x)$ and $\sqrt{Q_2} = (y, z^2)$. ■

Solution of (e). We set

$$\mathfrak{p} = (x),\ \mathfrak{q} = (y, z).$$

We then get

$$I_{\mathfrak{p}} = \left\{ \frac{r}{s} \,\middle|\, r \in I,\ s \notin \mathfrak{p} = (x) \right\},$$

and

$$I_{\mathfrak{q}} = \left\{ \frac{r}{s} \,\middle|\, r \in I,\ s \notin \mathfrak{q} = (y, z). \right\}$$

What we get is that

$$I_{\mathfrak{p}} = (x^2)R_{\mathfrak{p}}$$

and

$$I_{\mathfrak{q}} = (y, z^2)R_{\mathfrak{q}}.$$

Since $y \notin \mathfrak{p}$ we have

$$x^2 y y^{-1} = x^2 \in I_{\mathfrak{p}}$$

so

$$(x^2)R_{\mathfrak{p}} \subseteq I_{\mathfrak{p}}.$$

For the reverse inclusion notice that, as both generators of I are multiples of x^2, after localization every element generated by them is still in $(x^2)R_{\mathfrak{p}}$.

Together,

$$I_{\mathfrak{p}} = (x^2)R_{\mathfrak{p}}.$$

Then, similarly,

$$x^2 z^2 x^{-2} = z^2 \in I_{\mathfrak{q}}$$

and

$$x^2 y x^{-2} = y \in I_{\mathfrak{q}}$$

so the forward inclusion holds.

As before, we can see that it works out, or something.

■

Solution to (f). A point $p = (a, b, c) \in k^3$ lies in the affine variety $V(I)$ if and only if the generators are 0 on them,

$$a^2 b = 0,\ a^2 c^2 = 0.$$

If $a = 0$ both of these are 0 for any $b, c \in k$. If $a \neq 0$ we have that $b = 0$ and $c^2 = 0$ (and thus $c = 0$) since k is a field.

Thus

$$V(I) = \{(0, b, c) \mid b, c \in k\} \cup \{(a, 0, 0) \mid a \in k\}.$$

As $V(I) = V(\sqrt{I})$ and

$$I = Q_1 \cap Q_2$$

we have that

$$V(Q_1) = V(x)$$

and

$$V(Q_2) = V(y, z)$$

and these are reflected in what we have above since

$$V(x) = 0 \times k^2$$

and

$$V(y, z) = k \times 0 \times 0$$

as subsets of k^3. ■

Problem 3.2. Let

$$R = k[x, y],\ I = (x^2, xy),$$

and set

$$A = R/I.$$

(a) Prove that A is Noetherian. Determine whether A is Artinian. Give a useful normal form for elements of A.

(b) Find a primary decomposition of I. For every primary component Q_i, find $\sqrt{Q_i}$, determine whether Q_i is actually prime, and prove primaryness when it is not prime. Determine whether the decomposition is irredundant. Finally, determine whether the prime radicals occuring in the decompositio are comparable under inclusion.

(c) Compute $V(I) \subseteq k^2$ directly from generators. Then compute the varieties corresponding to the radicals from part (b).
Explain why the primary decomposition has more algebraic information than is visible from the irreducible compontents of $V(I)$.

(d) Let

$$\mathfrak{p} = (x),\ \mathfrak{m} = (x, y).$$

Compute $I_{\mathfrak{p}}$ and $I_{\mathfrak{m}}$.
For each primary component Q_i, determine whether $(Q_i)_{\mathfrak{p}}$ and $(Q_i)_{\mathfrak{m}}$ remain proper ideals or become the entire localized ring.

(e) Then consider the R-module $A = R/I$.

(f) Determine, for an arbitrary prime ideal $\mathfrak{q} \subseteq R$, exactly when $A_{\mathfrak{q}} = 0$.
Compare with $V(I)$.

(g) Suppose $I = Q_1 \cap Q_2$. Construct the maps and prove exactness of

$$0 \to R/I \to R/Q_1 \oplus R/Q_2 \to R/(Q_1 + Q_2) \to 0.$$

Then identify each of

$$R/Q_1,\ R/Q_2,\ R/(Q_1 + Q_2)$$

with familiar k-algebras and rewrite the sequence using those identifications.

(h) Inside

$$A = k[x, y]/(x^2, xy),$$

compute

$$\mathrm{Ann}_A(\overline{x}) \text{ and } \mathrm{Ann}_A(\overline{y}).$$

Take the inverse images of these annihilator ideals under $R \to A$.
Compare with the prime ideals in (b). Then use multiplication by $\overline{x}$,

$$\mu_x : A \to A,\ a \overset{\mu_x}{\mapsto} \overline{x}a,$$

to construct an exact sequence involving

$$\mathrm{Ann}_A(\overline{x}),\ A,\ A,\ A/(\overline{x}).$$

Identify $A/(\overline{x})$.

Proof of (a). There are several ways to prove Noetherianity. We can use that k is Noetherian so, by Hilbert's basis theorem, $k[x,y]$ is Noetherian.

Thus, since $I \subseteq R$ is a subideal of a Noetherian ring, I is Noetherian.

Then

$$A = R/I$$

is Noetherian.

However, A is not Artinian. Take

$$\overline{y} \supsetneq \overline{y^2} \supsetneq \overline{y^4} \supsetneq \cdots.$$

All elements of I are of the form

$$ax^2 + bxy = x(ax + by)$$

with

$$\overline{f} = q + x(ax + by) \in R.$$

Notice that if f is a multiple of x it is absorbed. ■

Solution of (b). We have

$$I = x(x,y) = (x) \cap (x,y)^2.$$

Then (x) is prime and thus primary while

$$\sqrt{(x,y)^2} = (x,y).$$

Then (x,y) is maximal since $k[x,y]/(x,y) \cong k$ and

$$(x,y)^2 \subseteq (x,y)^2 \subseteq (x,y)$$

so $(x,y)^2$ is primary. The decomposition is indeed irredundant since $x \notin (x,y)^2$ and $y^2 \notin (x)$.

The prime radicals are indeed comparable since $(x) \subseteq (x,y)$. ■

Solution of (c). We have

$$V(I) = V((x) \cap (x,y)^2) = V(x) \cup V((x,y)^2) = V(x) \cup V(x,y).$$

We have

$$V(x) = 0 \times k$$

and

$$V(x, y) = 0 \times 0$$

so

$$V(I) = 0 \times k.$$

With $I = (x) \cap (x, y)^2$ it is easier to see that $V(I) = 0 \times k$ than when working with (x^2, xy). ■

4 Exam style stuff

I have to practice for MAT224 Commutative Algebra oral examiniation so these problems are more about intuition, allowing less rigor.

Problem 4.1. Let

$$R = k[x, y],\ I = (x^2, xy, y^2),\ A = R/I.$$

What can you say about A? In particular, is it Noetherian or Artinian? Can you give a k-basis or normal form for its elements?
Compute $\sqrt{I}$. Then describe $V(I) \subseteq k^2$. What information about I is visible from $V(I)$?
Is I prime? Is I primary? Give the quickest convincing argument you can.
Determine the maximal ideals of A. Is A local? If so, identify the maximal ideal $\mathfrak{m}$, and compute enough powers of $\mathfrak{m}$ to see what happens.
Compute $\mathrm{Ann}_A(\bar{x})$ and $\mathrm{Ann}_A(\bar{y})$. Are either of these prime ideals?
Consider mutliplication by $\bar{x}$,

$$\mu_{\bar{x}} : A \to A,\ a \mapsto \bar{x}a.$$

Without being given a sequence, identify $\ker \mu_{\bar{x}}$, $\mathrm{im}\, \mu_{\bar{x}}$, $\mathrm{coker}\, \mu_{\bar{x}}$.

Solution. We have $I = (x, y)^2$. We can then say that the ring is Noetherian by using Hilbert's Basis Theorem twice $k \to k[x] \to k[x][y] = k[x, y]$ with Noetherian preserved. Then $I \subseteq R$ is Noetherian so $R/I = A$ is Noetherian.

Artinian does not hold for R since $(x) \supsetneq (x^2) \supsetneq (x^4) \supsetneq \cdots$ never stabilizes. We have an exact sequence

$$0 \to (x, y)(x, y) \to R = k[x, y] \to A \to 0.$$

We have that R is Artinian if and only if $(x, y)^2$ and A are Artinian. We know that R is not Artinian, and $(x, y)^2$ is not Artinian either so we can't conclude anything about A using this.

What we can notice is that any polynomials with x degree ≥ 2 vanish, same for y, while

terms in both x and y vanish also. Thus all polynomials have only constant terms, terms in x and terms in y. Thus A is an R-algebra, more specifically a k-vector space with basis $\{1, \bar{x}, \bar{y}\}$.

The root $\sqrt{I}$ is (x, y). We obviously have $(x, y) \subseteq \sqrt{I}$. For $ax + by \in (x, y)$ we have

$$(ax + by)^2 = a^2x^2 + (2ab)xy + b^2y^2$$

which is in I. Now, all generators of I are divisible by by either x or y so $I \subseteq (x, y)$. We have (x, y) prime (and even maximal) so $\sqrt{I} \subseteq (x, y)$ (since $f^n \in I \subseteq J$ implies $f \in J$ due to primality).

We do not have that I is prime, since x^2 is in I, but x is not. It is, however, maximal. We have already established that (x, y) is maximal and $I = (x, y)^2$ so

$$(x, y)^2 \subseteq I \subseteq (x, y).$$

Thus I is (x, y)-primal.

The maximal ideals of A map one to one with the maximal ideals of R containing I.

As mentioned we have $(x, y) \supseteq (x^2, xy, y^2)$ and (x, y) is maximal so

$$(\bar{x}, \bar{y}) = \frac{(x, y)}{(x^2, xy, y^2)}.$$

We can use the third isomorphism theorem to prove that $A/(\bar{x}, \bar{y}) \cong k$.

For uniqueness, suppose $f \in A$. Then $f = a + bx + cy$. The ideal $\mathfrak{m} = (\bar{x}, \bar{y})$ consists of all elements where the constant term is zero, $a = 0$, so an element f not in $(\bar{x}, \bar{y})$ is just $f = a \in k \setminus \{0\}$, which is invertible since k is a field.

As we glimpsed earlier, $\mathfrak{m}^2 = (\bar{x}, \bar{y})(\bar{x}, \bar{y}) = (\bar{x}^2, \bar{x}\bar{y}, \bar{y}^2) = 0$.

We have

$$\text{Ann}_A(\bar{x}) = \{a \in A \mid a\bar{x} = 0.\}$$

Equivalently, in $A = R/I$ we have that the annihilator of $\bar{x}$ are all $a = \alpha + \beta\bar{x} + \delta\bar{y}$ such that

$$a\bar{x} = \alpha\bar{x} + \underbrace{\beta\bar{x}^2}_{0} + \underbrace{\delta\bar{y}\bar{x}}_{0} = 0.$$

We have that $\bar{x}$ is a basiselement of the space, requiring $\alpha = 0$. β, δ can be any elements at all so

$$a = 0 + \beta\bar{x} + \delta\bar{y} \in (\bar{x}, \bar{y}) = \mathfrak{m}.$$

It is not terribly surprising that $\text{Ann}_A(\bar{x})$ is contained in $\mathfrak{m}$, since it is maximal, but actually $\text{Ann}_A(\bar{x}) = \mathfrak{m}$.

We can see this since, for $a\bar{x} + b\bar{y} \in (\bar{x}, \bar{y})$ we have

$$(a\bar{x} + b\bar{y})\bar{x} = a\bar{x}^2 + b\bar{y}^2 = 0.$$

Thus $\mathrm{Ann}_A(\bar{x}) = \mathfrak{m}$, the maximal ideal.

Now, the exact same holds for $\bar{y}$, for the same reasons, and it is not hard to see.

Now for multiplication at $\bar{x}$ we get a natural short exact sequence

$$0 \to A \xrightarrow{\mu_{\bar{x}}} A \xrightarrow{\pi} \mathrm{Ann}_A(\bar{x}) \to 0$$

■